Practical 11: Explicit and Implicit Intents

Source: MODULE 2.pdf | Source pages: 21–25

Aim

To develop and test an Android application demonstrating explicit and implicit intents.

Steps to Perform the Practical

1. Create a project named IntentDemo.

2. Add Explicit Intent and Implicit Intent buttons in activity_main.xml.

3. Create SecondActivity and its activity_second.xml layout.

4. In MainActivity.java, make the explicit button launch SecondActivity with an Intent.

5. Make the implicit button open a web URL using ACTION_VIEW and Uri.parse.

6. Run the app. Tap Explicit Intent to see the second screen, return, then tap Implicit Intent and choose a browser if prompted.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

Intents, Events and Listeners

4. Develop an Android application implementing Explicit and Implicit Intents.

ProjectName: IntentDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent" android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:id="@+id/main"
   android:padding="20dp">
   <Button
       android:id="@+id/btnExplicit"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Explicit Intent"/>
   <Button
       android:id="@+id/btnImplicit"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content" android:text="Implicit Intent"
       android:layout_marginTop="20dp"/>
</LinearLayout>
```

Create Second Activity.

Right Click: java package → com.example.projectname -> New→ Activity → Empty Views Activity

Activity Name: SecondActivity
Click: Finish

Design Second Activity Layout.
Open: res→ layout → activity_second.xml

File Name: activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:id="@+id/main"

   android:gravity="center">

   <TextView

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Welcome to Second Activity"

       android:textSize="24sp"/>

</LinearLayout>
```

File Name: MainActivity.java.

```
package com.example.intentdemo;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

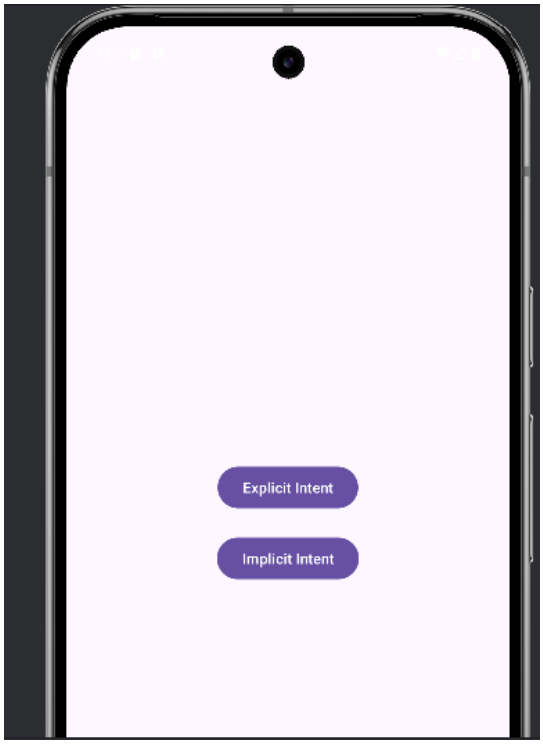
   Button btnExplicit, btnImplicit;

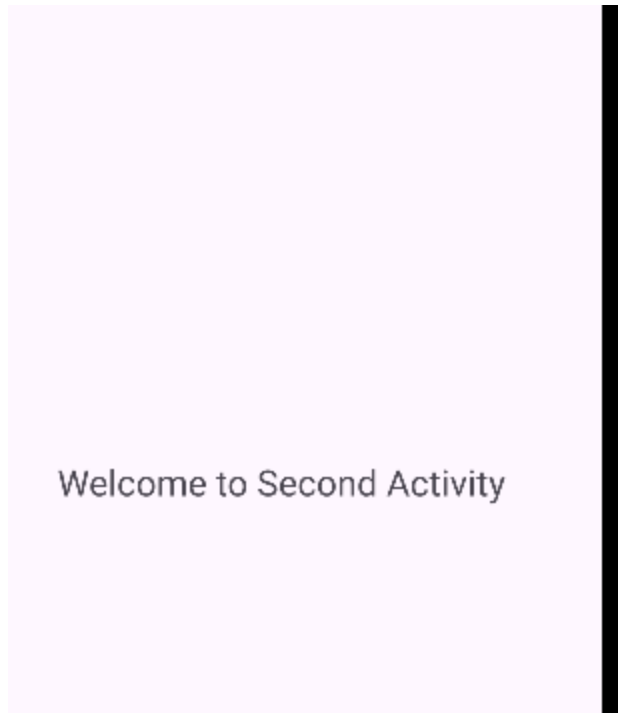
   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

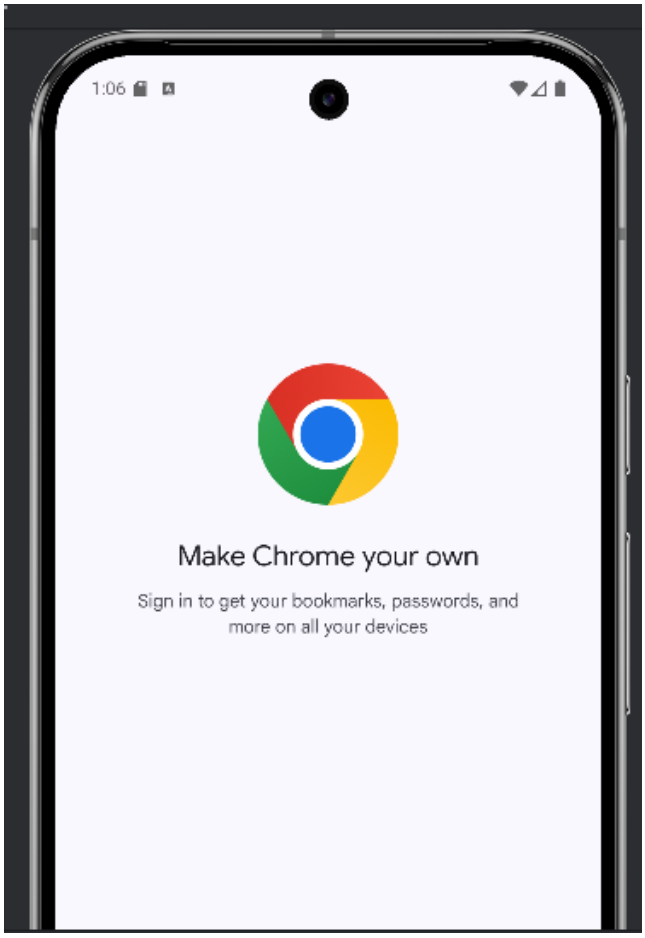
       btnExplicit = findViewById(R.id.btnExplicit);
       btnImplicit = findViewById(R.id.btnImplicit);

       // Explicit Intent
       btnExplicit.setOnClickListener(v -> {
           Intent intent = new Intent(MainActivity.this, SecondActivity.class);
           startActivity(intent);
       });
```

```
        // Implicit Intent
        btnImplicit.setOnClickListener(v -> {
            Intent intent = new Intent(
                    Intent.ACTION_VIEW,
                    Uri.parse("https://www.google.com")
            );
            startActivity(intent);
        });
    }
}
```


Explicit Intent
Implicit Intent


Welcome to Second Activity


1:06
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